



Arm talking blood pressure meter AD100

equipment and applications. The need for special skills to repair and maintain these equipment is realised as hospital technicians normally do not have the expertise and in many cases require special technical skill for rectification.

Quality, safety and reliability

Designing and manufacturing medical electronics equipment for compliance with product safety standards is no easy task. The best way to ascertain a product's compliance with its governing standard is to get type testing done on sample products by a third-party test house or recognised test lab after the design phase and before the product is launched in the market.

Once a newly designed product has undergone proper compliance testing, the dielectric withstand or hipot test is one of the primary safety tests routinely performed as part of the final production process. This test has been required on medical electronics products as well as most other electrical devices and appliances before they exit the manufacturer's production floor. The intention of the test is to stress a product's insulation beyond what it would encounter in normal use. The end goal is to ensure that patients or caregivers never serve as a current path to ground due to faulty insulation or faulty grounding within the product.

Medical device manufacturers' concerns are changing due to tighter governmental quality requirements,

regulations of agencies worldwide and current legal climates. When designing semiconductor products for medical original equipment manufacturers (OEMs), quality and reliability considerations are paramount and the stakes are high. Enhanced product flows to catalogue processes bring extended product lifetimes as well as well-defined and improved change-control processes.

To meet the demands of the medical market, dedicated and controlled manufacturing lines effectively eliminate facility-to-facility variations, extend qualification practices, improve product traceability, and enhance production testing rigours. Enhanced product flows can also save manufacturers cost and time-to-market by offering an alternative to upscrewing, which is commonplace in high-reliability markets. Another way to address these concerns is to adopt portions of ISO13485 (a quality management system) for medical devices as they apply to the semiconductor industry.

Expanding applications

With advancement in technology, new medical electronics devices are emerging in all healthcare segments. In hospitals these devices include handheld smart phone-sized ultrasound systems, digital stethoscopes and digital X-ray systems. Outside hospitals there are medical devices that ensure seamless vital sign monitoring with wireless connectiv-

ity between the hospital and home. In homes, available today are healthcare systems for daily activity monitoring, blood coagulation test, continuous glucose measurement and insulin delivery.

The latest medical systems deliver faster and more accurate diagnostic information, thus improving quality of life and reducing healthcare costs. A common requirement for all these devices is operation from a battery, small form factor, and a portable and lightweight design with intuitive user interface.

With devices like injectors or inhalers, the pharmaceutical sector is now looking for more electronics integration. This integration can allow it to develop new ways of administering drugs. Injected drugs can be inhaled and delivered through the lungs for improved efficiency, thanks to electronics-assisted devices.

Adding wireless connectivity allows doctors to remotely monitor and control the patient's compliance, preventing over-dosage. Automatic dose counting can also enable the automatic ordering of drugs from the pharmacy. Such promedical electronic advancements provide pharmaceutical companies with new differentiated benefits.

Medical electronics companies are developing innovative products that allow this industry to offer better quality care at reduced costs. Solutions like telemedicine help reduce the length of hospital stay and eliminate the need for frequent visits to a hospital. Early disease detection through modalities like ultrasound can significantly improve diagnosis.

Design challenges

Multi-disciplinary efforts have revolutionised medicine practices and equipment. The medical equipment should be safe, accurate and stable in data measurement, and efficient in emergency situations. Different systems need to be interoperable ensuring the highest efficiency without compromising on accuracy. Their embedded system should be small and compact and provide access to high-quality, affordable and accessible healthcare, anywhere cost-effectively.